

Integration Involving Partial fractions:

For integrating a rational function $\frac{P(x)}{Q(x)}$, there may be three cases, when $Q(x)$ has

Case I: Non repeated linear factors.

Case II: Repeated and non-repeated linear factors.

Case III: Linear and non-repeated irreducible quadratic factors or non-repeated irreducible quadratic factors.

EXERCISE 3.5

Evaluate the following integrals.

Q.1 $\int \frac{3x+1}{x^2-x-6} dx$

Solution:

$$\begin{aligned} & \frac{3x+1}{x^2-x-6} \\ &= \frac{3x+1}{x^2-3x+2x-6} \\ &= \frac{3x+1}{x(x-3)+2(x-3)} \\ &= \frac{3x+1}{(x-3)(x+2)} \end{aligned}$$

$$\text{Let } \frac{3x+1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2} \dots (i)$$

Multiplying both side by $(x-3)(x+2)$

$$3x+1 = A(x+2) + B(x-3) \dots (ii)$$

Put $x-3=0 \Rightarrow x=3$ in (ii)

$$3(3)+1 = A(3+2) + B(3-3)$$

$$10 = 5A \Rightarrow \boxed{A=2}$$

Put $x+2=0 \Rightarrow x=-2$ in (ii)

$$3(-2)+1 = A(-2+2) + B(-2-3)$$

$$-6+1 = -5B$$

$$-5B = -5 \Rightarrow \boxed{B=1}$$

Put $A=2$ and $B=1$ in (i)

$$\frac{3x+1}{(x-3)(x+2)} = \frac{2}{x-3} + \frac{1}{x+2}$$

Integrating both sides

$$\begin{aligned}\int \frac{3x+1}{(x-3)(x+2)} dx &= \int \frac{2}{x-3} dx + \int \frac{1}{x+2} dx \\ &= 2 \int \frac{1}{x-3} dx + \int \frac{1}{x+2} dx\end{aligned}$$

$$\int \frac{3x+1}{(x-3)(x+2)} dx = 2 \ln|x-3| + \ln|x+2| + c$$

Q.2 $\int \frac{5x+8}{(x+3)(2x-1)} dx$

Solution:

$$\text{Let } \frac{5x+8}{(x+3)(2x-1)} = \frac{A}{x+3} + \frac{B}{2x-1} \dots (i)$$

Multiplying both side by $(x+3)(2x-1)$

$$5x+8 = A(2x-1) + B(x+3) \dots (ii)$$

Put $x+3=0 \Rightarrow x=-3$ in (ii)

$$5(-3)+8 = A\{2(-3)-1\} + B(-3+3)$$

$$-15+8 = A(-6-1)$$

$$-7 = -7A \Rightarrow \boxed{A=1}$$

Put $2x-1=0 \Rightarrow x=\frac{1}{2}$ in (ii)

$$5\left(\frac{1}{2}\right)+8 = A\left\{2\left(\frac{1}{2}\right)-1\right\} + B\left(\frac{1}{2}+3\right)$$

$$\frac{5+16}{2} = 0 + B\left(\frac{1+6}{2}\right)$$

$$\frac{21}{2} = \frac{7B}{2} \Rightarrow \boxed{B=3}$$

Put $A=1$ and $B=3$ in (i)

$$\frac{5x+8}{(x+3)(2x-1)} = \frac{1}{x+3} + \frac{3}{2x-1}$$

Integrating both sides

$$\int \frac{5x+8}{(x+3)(2x-1)} dx = \int \frac{1}{x+3} dx + 3 \int \frac{1}{2x-1} dx$$

$$\int \frac{5x+8}{(x+3)(2x-1)} dx = \int \frac{1}{x+3} dx + \frac{3}{2} \int \frac{2}{2x-1} dx$$

$$\int \frac{5x+8}{(x+3)(2x-1)} dx = \ln|x+3| + \frac{3}{2} \ln|2x-1| + c$$

Q.3 $\int \frac{x^2 + 3x - 34}{x^2 + 2x - 15} dx$

Solution:

By long division

$$\begin{array}{r} x^2 + 2x - 15 \overline{) \begin{array}{c} x^2 + 3x - 34 \\ \underline{+x^2 + 2x - 15} \\ x - 19 \end{array}} \end{array}$$

So

$$\begin{aligned} \frac{x^2 + 3x - 34}{x^2 + 2x - 15} &= 1 + \frac{x - 19}{x^2 + 2x - 15} \\ &= 1 + \frac{x - 19}{x^2 + 2x - 15} \\ &= 1 + \frac{x - 19}{x^2 + 5x - 3x - 15} \\ &= 1 + \frac{x - 19}{x(x+5) - 3(x+5)} \\ &= 1 + \frac{x - 19}{(x+5)(x-3)} \quad \dots(i) \end{aligned}$$

$$\text{Let } \frac{x-19}{(x+5)(x-3)} = \frac{A}{x+5} + \frac{B}{x-3} \quad \dots(ii)$$

Multiplying both sides by $(x+5)(x-3)$

$$x-19 = A(x-3) + B(x+5) \quad \dots(iii)$$

Put $x+5=0 \Rightarrow x=-5$ in (iii)

$$-5-19 = A(-5-3) + B(-5+5)$$

$$-24 = A(-8) \Rightarrow \boxed{A=3}$$

Put $x-3=0 \Rightarrow x=3$ in (iii)

$$3-19 = A(3-3) + B(3+5)$$

$$-16 = 8B \Rightarrow \boxed{B=-2}$$

Put $A=3$ and $B=-2$ in (ii)

$$\frac{x-19}{(x+5)(x-3)} = \frac{3}{x+5} - \frac{2}{x-3}$$

(i) becomes

$$\frac{x^2 + 3x - 34}{x^2 + 2x - 15} = 1 + \frac{3}{x+5} - \frac{2}{x-3}$$

Integrating both sides

$$\int \frac{x^2 + 3x - 34}{x^2 + 2x - 15} dx = \int 1 dx + 3 \int \frac{1}{x+5} dx - 2 \int \frac{1}{x-3} dx$$

$$\int \frac{x^2 + 3x - 34}{x^2 + 2x - 15} dx = x + 3 \ln|x+5| - 2 \ln|x-3| + c$$

Q.4 $\int \frac{(a-b)x}{(x-a)(x-b)} dx, (a > b)$

Solution:

Let $\frac{(a-b)x}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b} \dots\dots(i)$

Multiplying both sides by $(x-a)(x-b)$

$$(a-b)x = A(x-b) + B(x-a) \dots\dots(ii)$$

Put $x-a=0 \Rightarrow x=a$ in (ii)

$$(a-b)a = A(a-b) + 0$$

$$(a-b)a = A(a-b) \Rightarrow \boxed{A=a}$$

Put $x-b=0 \Rightarrow x=b$ in (ii)

$$(a-b)(b) = 0 + B(b-a)$$

$$(a-b)(b) = -B(a-b) \Rightarrow \boxed{B=-b}$$

Put $A=a$ and $B=-b$ in (i)

$$\frac{(a-b)x}{(x-a)(x-b)} = \frac{a}{x-a} - \frac{b}{x-b}$$

Integrating both sides

$$\begin{aligned} \int \frac{(a-b)x}{(x-a)(x-b)} dx &= \int \frac{a}{x-a} dx - \int \frac{b}{x-b} dx \\ &= a \int \frac{1}{x-a} dx - b \int \frac{1}{x-b} dx \\ &= a \ln|x-a| - b \ln|x-b| + c \end{aligned}$$

Q.5 $\int \frac{3-x}{1-x-6x^2} dx$

Solution:

$$\begin{aligned} \frac{3-x}{1-x-6x^2} &= \frac{-(x-3)}{-(6x^2+x-1)} \\ \frac{x-3}{6x^2+x-1} &= \frac{x-3}{6x^2+3x-2x-1} \\ &= \frac{x-3}{3x(2x+1)-1(2x+1)} \end{aligned}$$

$$= \frac{x-3}{(2x+1)(3x-1)}$$

$$\text{Let } \frac{x-3}{(2x+1)(3x-1)} = \frac{A}{2x+1} + \frac{B}{3x-1} \dots (i)$$

Multiplying both sides by $(2x+1)(3x-1)$

$$x-3 = A(3x-1) + B(2x+1) \dots (ii)$$

$$\text{Put } 2x+1=0 \Rightarrow x = -\frac{1}{2} \text{ in (ii)}$$

$$-\frac{1}{2}-3 = A\left\{3\left(\frac{-1}{2}\right)-1\right\} + B\left\{2\left(\frac{-1}{2}\right)+1\right\}$$

$$\frac{-1-6}{2} = A\left(\frac{-3-2}{2}\right) + B(-1+1)$$

$$\frac{-7}{2} = A\left(\frac{-5}{2}\right) \Rightarrow \boxed{A = \frac{7}{5}}$$

$$\text{Put } 3x-1=0 \Rightarrow x = \frac{1}{3} \text{ in (ii)}$$

$$\frac{1}{3}-3 = A\left\{3\left(\frac{1}{3}\right)-1\right\} + B\left\{2\left(\frac{1}{3}\right)+1\right\}$$

$$\frac{1-9}{3} = B\left(\frac{2+3}{3}\right)$$

$$\frac{-8}{3} = B\left(\frac{5}{3}\right) \Rightarrow \boxed{B = \frac{-8}{5}}$$

$$\text{Put } A = \frac{7}{5} \text{ and } B = \frac{-8}{5} \text{ in eq. (i)}$$

$$\frac{x-3}{(2x+1)(3x-1)} = \frac{\left(\frac{7}{5}\right)}{2x+1} + \frac{\left(\frac{-8}{5}\right)}{3x-1}$$

$$\frac{3-x}{1-x-6x^2} = \frac{7}{5(2x+1)} - \frac{8}{5(3x-1)}$$

Integrating both sides

$$\begin{aligned} \int \frac{3-x}{1-x-6x^2} dx &= \frac{7}{5} \int \frac{1}{2x+1} dx - \frac{8}{5} \int \frac{1}{3x-1} dx \\ &= \frac{7}{5 \times 2} \int \frac{2}{2x+1} dx - \frac{8}{5 \times 3} \int \frac{3}{3x-1} dx \\ &= \frac{7}{10} \ln|2x+1| - \frac{8}{15} \ln|3x-1| + c \end{aligned}$$

Q.6 $\int \frac{2x}{x^2 - a^2} dx, (x > a)$

Solution:

Consider $\frac{2x}{x^2 - a^2} = \frac{2x}{(x-a)(x+a)}$

Let $\frac{2x}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a} \dots$ (i)

Multiplying both sides by $(x-a)(x+a)$

$2x = A(x+a) + B(x-a) \dots$ (ii)

Put $x-a=0 \Rightarrow x=a$ in (ii)

$2a = A(a+a) + B(a-a)$

$2a = 2aA \Rightarrow \boxed{A=1}$

Put $x+a=0 \Rightarrow x=-a$ in (ii)

$2(-a) = A(-a+a) + B(-a-a)$

$-2a = -2aB \Rightarrow \boxed{B=1}$

Put $A=1$ and $B=1$ in eq. (i)

$\frac{2x}{x^2 - a^2} = \frac{1}{x-a} + \frac{1}{x+a}$

Integrating both sides

$$\begin{aligned} \int \frac{2x}{x^2 - a^2} dx &= \int \frac{1}{x-a} dx + \int \frac{1}{x+a} dx \\ &= \ln|x-a| + \ln|x+a| + c \\ &= \ln|(x-a)(x+a)| + c \end{aligned}$$

$\int \frac{2x}{x^2 - a^2} dx = \ln|x^2 - a^2| + c$

Alternate Solution:

$$\begin{aligned} &\int \frac{2x}{x^2 - a^2} dx \\ &= \ln(x^2 - a^2) + c \quad \because \int \frac{f'(x)}{f(x)} dx = \ln f(x) + c \end{aligned}$$

Q.7 $\int \frac{1}{6x^2 + 5x - 4} dx$

Solution:

$$\begin{aligned} \frac{1}{6x^2 + 5x - 4} &= \frac{1}{6x^2 + 8x - 3x - 4} \\ &= \frac{1}{2x(3x+4) - 1(3x+4)} = \frac{1}{(3x+4)(2x-1)} \end{aligned}$$

Let $\frac{1}{(3x+4)(2x-1)} = \frac{A}{3x+4} + \frac{B}{2x-1} \dots$ (i)

Multiplying both sides by $(3x+4)(2x-1)$

$$1 = A(2x-1) + B(3x+4) \dots (ii)$$

$$\text{Put } 3x+4=0 \Rightarrow x = \frac{-4}{3} \text{ in (ii)}$$

$$1 = A \left\{ 2 \left(\frac{-4}{3} \right) - 1 \right\} + B \left\{ 3 \left(\frac{-4}{3} \right) + 4 \right\}$$

$$1 = A \left\{ \frac{-8-3}{3} \right\}$$

$$1 = A \left(\frac{-11}{3} \right) \Rightarrow \boxed{A = \frac{-3}{11}}$$

$$\text{Put } 2x-1=0 \Rightarrow x = \frac{1}{2} \text{ in (ii)}$$

$$1 = A \left\{ 2 \left(\frac{1}{2} \right) - 1 \right\} + B \left\{ 3 \left(\frac{1}{2} \right) + 4 \right\}$$

$$1 = B \left(\frac{3+8}{2} \right)$$

$$1 = B \left(\frac{11}{2} \right) \Rightarrow \boxed{B = \frac{2}{11}}$$

$$\text{Put } A = \frac{-3}{11} \text{ and } B = \frac{2}{11} \text{ in eq. (i)}$$

$$\frac{1}{(3x+4)(2x-1)} = \frac{\left(\frac{-3}{11} \right)}{3x+4} + \frac{\left(\frac{2}{11} \right)}{2x-1}$$

$$\frac{1}{6x^2+5x-4} = \frac{-3}{11(3x+4)} + \frac{2}{11(2x-1)}$$

Integrating both sides

$$\begin{aligned} \int \frac{1}{6x^2+5x-4} dx &= \frac{-1}{11} \int \frac{3}{3x+4} dx + \frac{1}{11} \int \frac{2}{2x-1} dx \\ &= \frac{-1}{11} \ln|3x+4| + \frac{1}{11} \ln|2x-1| + c \\ &= \frac{1}{11} (\ln|2x-1| - \ln|3x+4|) + c \end{aligned}$$

$$\int \frac{1}{6x^2+5x-4} dx = \frac{1}{11} \ln \left| \frac{2x-1}{3x+4} \right| + c$$

Q.8 $\int \frac{2x^3 - 3x^2 - x - 7}{2x^2 - 3x - 2} dx$

Solution:

By long division

$$\begin{array}{r}
 2x^2 - 3x - 2 \overline{) \begin{array}{c} x \\ 2x^3 - 3x^2 - x - 7 \\ \underline{+2x^3 + 3x^2 + 2x} \\ x - 7 \end{array}} \\
 \frac{2x^3 - 3x^2 - x - 7}{2x^2 - 3x - 2} = x + \frac{x-7}{2x^2 - 3x - 2} \\
 = x + \frac{x-7}{2x^2 - 4x + x - 2} \\
 = x + \frac{x-7}{2x(x-2) + 1(x-2)} \\
 = x + \frac{x-7}{(x-2)(2x+1)} \dots (i)
 \end{array}$$

$$\text{Let } \frac{x-7}{(x-2)(2x+1)} = \frac{A}{x-2} + \frac{B}{2x+1} \dots (ii)$$

Multiplying both sides by $(x-2)(2x+1)$

$$x-7 = A(2x+1) + B(x-2) \dots (iii)$$

Put $x-2=0 \Rightarrow x=2$ in (iii)

$$2-7 = A\{2(2)+1\} + B(2-2)$$

$$-5 = 5A \Rightarrow \boxed{A=-1}$$

Put $2x+1=0 \Rightarrow x = \frac{-1}{2}$ in (iii)

$$-\frac{1}{2} - 7 = A\left\{2\left(\frac{-1}{2}\right) + 1\right\} + B\left(\frac{-1}{2} - 2\right)$$

$$\frac{-15}{2} = 0 + B\left(\frac{-5}{2}\right) \Rightarrow \boxed{B=3}$$

Put $A=-1$ and $B=3$ in (ii)

$$\frac{x-7}{(x-2)(2x+1)} = \frac{-1}{x-2} + \frac{3}{2x+1}$$

(i) becomes

$$\frac{2x^3 - 3x^2 - x - 7}{2x^2 - 3x - 2} = x - \frac{1}{x-2} + \frac{3}{2x+1}$$

Integrating both sides

$$\begin{aligned}
 \int \frac{2x^3 - 3x^2 - x - 7}{2x^2 - 3x - 2} dx &= \int x dx - \int \frac{1}{x-2} dx + \frac{3}{2} \int \frac{2}{2x+1} dx \\
 &= \frac{x^2}{2} - \ln|x-2| + \frac{3}{2} \ln|2x+1| + c
 \end{aligned}$$

Q.9 $\int \frac{3x^2 - 12x + 11}{(x-1)(x-2)(x-3)} dx$

Solution:

$$\text{Let } \frac{3x^2 - 12x + 11}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} \dots (i)$$

Multiplying both sides by $(x-1)(x-2)(x-3)$

$$3x^2 - 12x + 11 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2) \dots (ii)$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$3(1)^2 - 12(1) + 11 = A(1-2)(1-3)$$

$$2 = A(-1)(-2) \Rightarrow \boxed{A=1}$$

Put $x-2=0 \Rightarrow x=2$ in (ii)

$$3(2)^2 - 12(2) + 11 = B(2-1)(2-3)$$

$$12 - 24 + 11 = B(1)(-1)$$

$$-1 = B(-1) \Rightarrow \boxed{B=1}$$

Put $x-3=0 \Rightarrow x=3$ in (ii)

$$3(3)^2 - 12(3) + 11 = (3-1)(3-2)$$

$$27 - 36 + 11 = C(2)(1)$$

$$2 = 2C \Rightarrow \boxed{C=1}$$

Put $A=1, B=1, C=1$ in (i)

$$\frac{3x^2 - 12x + 11}{(x-1)(x-2)(x-3)} = \frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x-3}$$

Integrating both sides

$$\begin{aligned} \int \frac{3x^2 - 12x + 11}{(x-1)(x-2)(x-3)} dx &= \int \frac{1}{x-1} dx + \int \frac{1}{x-2} dx + \int \frac{1}{x-3} dx \\ &= \ln|x-1| + \ln|x-2| + \ln|x-3| + c \end{aligned}$$

Q.10 $\int \frac{2x-1}{x(x-1)(x-3)} dx$

Solution:

$$\text{Let } \frac{2x-1}{x(x-1)(x-3)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x-3} \dots (i)$$

Multiplying both sides by $x(x-1)(x-3)$

$$2x-1 = A(x-1)(x-3) + Bx(x-3) + Cx(x-1) \dots (ii)$$

Put $x=0$ in (ii)

$$2(0)-1 = A(0-1)(0-3)$$

$$-1 = A(-1)(-3) \Rightarrow \boxed{A = -\frac{1}{3}}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$2(1)-1=B(1)(1-3)$$

$$1=B(-2) \Rightarrow \boxed{B=-\frac{1}{2}}$$

Put $x-3=0 \Rightarrow x=3$ in (ii)

$$2(3)-1=C(3)(3-1)$$

$$5=C(3)(2) \Rightarrow \boxed{C=\frac{5}{6}}$$

Put $A=\frac{-1}{3}, B=\frac{-1}{2}$ and $C=\frac{5}{6}$ in (i)

$$\begin{aligned} \frac{2x-1}{x(x-1)(x-3)} &= \frac{\left(\frac{-1}{3}\right)}{x} + \frac{\left(\frac{-1}{2}\right)}{x-1} + \frac{\left(\frac{5}{6}\right)}{x-3} \\ \frac{2x-1}{x(x-1)(x-3)} &= \frac{-1}{3x} - \frac{1}{2(x-1)} + \frac{5}{6(x-3)} \end{aligned}$$

Integrating both sides

$$\begin{aligned} \int \frac{2x-1}{x(x-1)(x-3)} dx &= \frac{-1}{3} \int \frac{1}{x} dx - \int \frac{1}{2(x-1)} dx + \frac{5}{6} \int \frac{1}{x-3} dx \\ &= \frac{-1}{3} \int \frac{1}{x} dx - \frac{1}{2} \int \frac{1}{x-1} dx + \frac{5}{6} \int \frac{1}{x-3} dx \\ &= \frac{-1}{3} \ln|x| - \frac{1}{2} \ln|x-1| + \frac{5}{6} \ln|x-3| + c \end{aligned}$$

Q.11 $\int \frac{5x^2+9x+6}{(x^2-1)(2x+3)} dx$

Solution:

$$\frac{5x^2+9x+6}{(x^2-1)(2x+3)} = \frac{5x^2+9x+6}{(x-1)(x+1)(2x+3)}$$

$$\text{Let } \frac{5x^2+9x+6}{(x-1)(x+1)(2x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{2x+3} \dots \text{ (i)}$$

Multiplying both sides by $(x-1)(x+1)(2x+3)$

$$5x^2+9x+6=A(x+1)(2x+3)+B(x-1)(2x+3)+C(x-1)(x+1) \dots \text{ (ii)}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$5(1)^2+9(1)+6=A(1+1)\{2(1)+3\}$$

$$20=A(2)(5) \Rightarrow \boxed{A=2}$$

Put $x+1=0 \Rightarrow x=-1$ in (ii)

$$5(-1)^2 + 9(-1) + 6 = B(-1-1)\{2(-1)+3\}$$

$$11-9 = B(-2)(1)$$

$$2 = -2B \Rightarrow \boxed{B = -1}$$

$$\text{Put } 2x+3=0 \Rightarrow x = \frac{-3}{2} \text{ in (ii)}$$

$$5\left(\frac{-3}{2}\right)^2 + 9\left(\frac{-3}{2}\right) + 6 = C\left(\frac{-3}{2}-1\right)\left(\frac{-3}{2}+1\right)$$

$$5\left(\frac{9}{4}\right) - \frac{27}{2} + 6 = C\left(\frac{-3-2}{2}\right)\left(\frac{-3+2}{2}\right)$$

$$\frac{45-54+24}{4} = C\left(\frac{-5}{2}\right)\left(\frac{-1}{2}\right)$$

$$\frac{15}{4} = C\left(\frac{5}{4}\right) \Rightarrow \boxed{C=3}$$

$$\text{Put } A=2, B=-1, C=3 \text{ in (i)}$$

$$\frac{5x^2+9x+6}{(x^2-1)(2x+3)} = \frac{2}{x-1} - \frac{1}{x+1} + \frac{3}{2x+3}$$

Integrating both sides

$$\begin{aligned} \int \frac{5x^2+9x+6}{(x^2-1)(2x+3)} dx &= 2 \int \frac{1}{x-1} dx - \int \frac{1}{x+1} dx + 3 \int \frac{1}{2x+3} dx \\ &= 2 \int \frac{1}{x-1} dx - \int \frac{1}{x+1} dx + \frac{3}{2} \int \frac{2}{2x+3} dx \\ &= 2 \ln|x-1| - \ln|x+1| + \frac{3}{2} \ln|2x+3| + c \end{aligned}$$

Q.12 $\int \frac{4+7x}{(1+x)^2(2x+3)} dx$

Solution:

$$\text{Let } \frac{4+7x}{(1+x)^2(2+3x)} = \frac{A}{2+3x} + \frac{B}{1+x} + \frac{C}{(1+x)^2} \dots \text{(i)}$$

Multiplying both sides by $(1+x)^2(2+3x)$

$$4+7x = A(1+x)^2 + B(1+x)(2+3x) + C(2+3x) \dots \text{(ii)}$$

$$\text{Put } 2+3x=0 \Rightarrow x = \frac{-2}{3} \text{ in (ii)}$$

$$4+7\left(\frac{-2}{3}\right) = A\left(1-\frac{2}{3}\right)^2$$

$$\frac{12-14}{3} = A\left(\frac{3-2}{3}\right)^2$$

$$\frac{-2}{3} = A\left(\frac{1}{9}\right)$$

$$A = \frac{-2}{3} \times 9 \Rightarrow \boxed{A = -6}$$

Put $1+x=0 \Rightarrow x=-1$ in (ii)

$$4+7(-1) = C\{2+3(-1)\}$$

$$4-7 = C(2-3)$$

$$-3 = C(-1) \Rightarrow \boxed{C=3}$$

From (ii)

$$4+7x = A(1+x^2+2x) + B(2+3x+2x+3x^2) + C(2+3x)$$

$$4+7x = A + Ax^2 + 2Ax + 2B + 5Bx + 3Bx^2 + 2C + 3Cx$$

Comparing the coefficients of x^2

$$0 = A + 3B$$

$$0 = -6 + 3B$$

$$3B = 6 \Rightarrow \boxed{B=2}$$

Put $A=-6, B=2$ and $C=3$ in (i)

$$\frac{4+7x}{(1+x)^2(2+3x)} = \frac{-6}{2+3x} + \frac{2}{1+x} + \frac{3}{(1+x)^2}$$

Integrating both sides

$$\begin{aligned} \int \frac{4+7x}{(1+x)^2(2+3x)} dx &= -2 \int \frac{3}{2+3x} dx + 2 \int \frac{1}{1+x} dx + 3 \int (1+x)^{-2} dx \\ &= -2 \ln|2+3x| + 2 \ln|1+x| + 3 \frac{(1+x)^{-2+1}}{(-2+1)} + c \\ &= -2 \ln|2+3x| + 2 \ln|1+x| - \frac{3}{1+x} + c \end{aligned}$$

Q.13 $\int \frac{2x^2}{(x-1)^2(x+1)} dx$

Solution:

$$\text{Let } \frac{2x^2}{(x-1)^2(x+1)} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \dots \text{ (i)}$$

Multiplying both side by $(x-1)^2(x+1)$

$$2x^2 = A(x-1)^2 + B(x-1)(x+1) + C(x+1) \dots \text{ (ii)}$$

Put $x+1=0 \Rightarrow x=-1$ in (ii)

$$2(-1)^2 = A(-1-1)^2$$

$$2 = A(-2)^2$$

$$2 = 4A \Rightarrow \boxed{A = \frac{1}{2}}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$2(1)^2 = C(1+1)$$

$$2 = 2C \Rightarrow \boxed{C=1}$$

From (ii)

$$2x^2 = A(x^2 + 1 - 2x) + B(x^2 - 1) + C(x + 1)$$

$$2x^2 = Ax^2 + A - 2Ax + Bx^2 - B + Cx + C$$

Comparing the coefficients of ' x^2 '

$$2 = A + B$$

$$2 = \frac{1}{2} + B$$

$$B = 2 - \frac{1}{2} \Rightarrow \boxed{B = \frac{3}{2}}$$

Put $A = \frac{1}{2}$, $B = \frac{3}{2}$ and $C = 1$ in (i)

$$\frac{2x^2}{(x-1)^2(x+1)} = \frac{1}{2(x+1)} + \frac{3}{2(x-1)} + \frac{1}{(x-1)^2}$$

Integrating both sides

$$\begin{aligned} \int \frac{2x^2}{(x-1)^2(x+1)} dx &= \frac{1}{2} \int \frac{1}{x+1} dx + \frac{3}{2} \int \frac{1}{x-1} dx + \int (x-1)^{-2} dx \\ &= \frac{1}{2} \ln|x+1| + \frac{3}{2} \ln|x-1| + \frac{(x-1)^{-2+1}}{-2+1} + c \\ &= \frac{1}{2} \ln|x+1| + \frac{3}{2} \ln|x-1| - (x-1)^{-1} + c \\ &= \frac{1}{2} \ln|x+1| + \frac{3}{2} \ln|x-1| - \frac{1}{x-1} + c \end{aligned}$$

Q.14 $\int \frac{1}{(x-1)(x+1)^2} dx$

Solution:

$$\text{Let } \frac{1}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2} \dots \text{(i)}$$

Multiplying both side by $(x-1)(x+1)^2$

$$1 = A(x+1)^2 + B(x-1)(x+1) + C(x-1) \dots \text{(ii)}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$1 = A(1+1)^2$$

$$1 = 4A \Rightarrow \boxed{A = \frac{1}{4}}$$

Put $x+1=0 \Rightarrow x=-1$ in (ii)

$$1 = C(-1-1)$$

$$1 = C(-2) \Rightarrow \boxed{C = -\frac{1}{2}}$$

Expanding (ii)

$$1 = A(x^2 + 1 + 2x) + B(x^2 - 1) + C(x - 1)$$

$$1 = Ax^2 + A + 2Ax + Bx^2 - B + Cx - C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = \frac{1}{4} + B \Rightarrow \boxed{B = -\frac{1}{4}}$$

Put $A = \frac{1}{4}, B = -\frac{1}{4}$ and $C = \frac{-1}{2}$ in (i)

$$\frac{1}{(x-1)(x+1)^2} = \frac{1}{4(x-1)} - \frac{1}{4(x+1)} - \frac{1}{2(x+1)^2}$$

Integrating both sides

$$\begin{aligned} \int \frac{1}{(x-1)(x+1)^2} dx &= \int \frac{1}{4(x-1)} dx - \frac{1}{4} \int \frac{1}{x+1} dx - \frac{1}{2} \int (x+1)^{-2} dx \\ &= \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| - \frac{1}{2} \frac{(x+1)^{-1}}{(-1)} + c \\ &= \frac{1}{4} \ln|x-1| - \frac{1}{4} \ln|x+1| + \frac{1}{2(x+1)} + c \end{aligned}$$

Q.15 $\int \frac{x+4}{x^3-3x^2+4} dx$

Solution:

First we factorize $x^3 - 3x^2 + 4$

Clearly $(x+1)$ is its factor

Using synthetic division

-1	1	-3	0	4
		-1	4	-4
	1	-4	4	0

Quotient = $x^2 - 4x + 4$

$$\begin{aligned} x^3 - 3x^2 + 4 &= (x+1)(x^2 - 4x + 4) \\ &= (x+1)(x-2)^2 \end{aligned}$$

$$\frac{x+4}{x^3-3x^2+4} = \frac{x+4}{(x+1)(x-2)^2}$$

Let $\frac{x+4}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2} \dots (i)$

Multiplying both sides by $(x+1)(x-2)^2$

$$x+4 = A(x-2)^2 + B(x+1)(x-2) + C(x+1) \dots (ii)$$

Put $x+1=0 \Rightarrow x=-1$ in (ii)

$$-1+4 = A(-1-2)^2$$

$$3 = 9A \Rightarrow \boxed{A = \frac{1}{3}}$$

Put $x-2=0 \Rightarrow x=2$

Put $x=2$ in (ii)

$$2+4 = C(2+1)$$

$$6 = 3C \Rightarrow \boxed{C = 2}$$

Expanding (ii)

$$x + 4 = A(x^2 + 4 - 4x) + B(x^2 - x - 2) + C(x + 1)$$

$$x + 4 = Ax^2 + 4A - 4Ax + Bx^2 - Bx - 2B + Cx + C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = \frac{1}{3} + B \Rightarrow \boxed{B = -\frac{1}{3}}$$

Put $A = \frac{1}{3}, B = -\frac{1}{3}$ and $C = 2$ in (i)

$$\frac{x+4}{(x+1)(x-2)^2} = \frac{\left(\frac{1}{3}\right)}{x+1} + \frac{\left(-\frac{1}{3}\right)}{x-2} + \frac{2}{(x-2)^2}$$

$$\frac{x+4}{x^3-3x^2+4} = \frac{1}{3(x+1)} - \frac{1}{3(x-2)} + \frac{2}{(x-2)^2}$$

Integrating both sides

$$\begin{aligned} \int \frac{x+4}{x^3-3x^2+4} dx &= \frac{1}{3} \int \frac{1}{x+1} dx - \frac{1}{3} \int \frac{1}{x-2} dx + 2 \int (x-2)^{-2} dx \\ &= \frac{1}{3} \ln|x+1| - \frac{1}{3} \ln|x-2| + 2 \frac{(x-2)^{-2+1}}{(-2+1)} + c \\ &= \frac{1}{3} \ln|x+1| - \frac{1}{3} \ln|x-2| - \frac{2}{x-2} + c \end{aligned}$$

Q.16 $\int \frac{x^3 - 6x^2 + 25}{(x+1)^2(x-2)^2} dx$

Solution:

$$\text{Let } \frac{x^3 - 6x^2 + 25}{(x+1)^2(x-2)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-2} + \frac{D}{(x-2)^2} \dots (i)$$

Multiplying both sides by $(x+1)^2(x-2)^2$

$$x^3 - 6x^2 + 25 = A(x+1)(x-2)^2 + B(x-2)^2 + C(x+1)^2(x-2) + D(x+1)^2 \dots (ii)$$

Put $x+1=0 \Rightarrow x=-1$ in (ii)

$$(-1)^3 - 6(-1)^2 + 25 = B(-1-2)^2$$

$$-1 - 6 + 25 = B(-3)^2$$

$$18 = 9B \Rightarrow \boxed{B = 2}$$

Put $x-2=0 \Rightarrow x=2$ in (ii)

$$(2)^3 - 6(2)^2 + 25 = D(2+1)^2$$

$$8 - 6(4) + 25 = D(3)^2$$

$$9 = 9D \Rightarrow \boxed{D = 1}$$

Expanding (ii)

$$\begin{aligned}
 x^3 - 6x^2 + 25 &= A(x+1)(x^2 + 4 - 4x) + B(x^2 + 4 - 4x) + C(x^2 + 1 + 2x)(x-2) + D(x^2 + 1 + 2x) \\
 x^3 - 6x^2 + 25 &= A(x^3 + 4x - 4x^2 + x^2 + 4 - 4x) + B(x^2 - 4x + 4) + C(x^3 - 2x^2 + x - 2 + 2x^2 - 4x) \\
 &\quad + D(x^2 + 1 + 2x)
 \end{aligned}$$

$$x^3 - 6x^2 + 25 = Ax^3 - 3Ax^2 + 4A + Bx^2 - 4Bx + 4B + Cx^3 - 3Cx - 2C + Dx^2 + D + 2Dx$$

Comparing the coefficients of x^3 and x^2 respectively

$$1 = A + C \quad \dots \text{ (iii)}$$

$$-6 = -3A + B + D \quad \dots \text{ (iv)}$$

Put $B = 2$ and $D = 1$ in (iv)

$$-6 = -3A + 2 + 1$$

$$-6 = -3A + 3$$

$$3A = 6 + 3$$

$$3A = 9 \Rightarrow \boxed{A = 3}$$

Put $A = 3$ in (iii)

$$1 = 3 + C$$

$$C = 1 - 3 \Rightarrow \boxed{C = -2}$$

Put $A = 3, B = 2, C = -2, D = 1$ in (i)

$$\frac{x^3 - 6x^2 + 25}{(x+1)^2(x-2)^2} = \frac{3}{x+1} + \frac{2}{(x+1)^2} - \frac{2}{x-2} + \frac{1}{(x-2)^2}$$

Integrating both sides

$$\begin{aligned}
 \int \frac{x^3 - 6x^2 + 25}{(x+1)^2(x-2)^2} dx &= 3 \int \frac{1}{x+1} dx + 2 \int (x+1)^{-2} dx - 2 \int \frac{1}{x-2} dx + \int (x-2)^{-2} dx \\
 &= 3 \ln|x+1| + 2 \frac{(x+1)^{-2+1}}{(-2+1)} - 2 \ln|x-2| + \frac{(x-2)^{-2+1}}{(-2+1)} + c \\
 &= 3 \ln|x+1| - \frac{2}{x+1} - 2 \ln|x-2| - \frac{1}{x-2} + c
 \end{aligned}$$

Q.17 $\int \frac{x^3 + 22x^2 + 14x - 17}{(x-3)(x+2)^3} dx$

Solution:

$$\text{Let } \frac{x^3 + 22x^2 + 14x - 17}{(x-3)(x+2)^3} = \frac{A}{x-3} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3} \quad \dots \text{ (i)}$$

Multiplying both sides by $(x-3)(x+2)^3$

$$x^3 + 22x^2 + 14x - 17 = A(x+2)^3 + B(x-3)(x+2)^2 + C(x-3)(x+2) + D(x-3) \quad \dots \text{ (ii)}$$

Put $x-3=0 \Rightarrow x=3$ in (ii)

$$(3)^3 + 22(3)^2 + 14(3) - 17 = A(3+2)^3$$

$$27 + 198 + 42 - 17 = A(5)^3$$

$$250 = 125A \Rightarrow \boxed{A = 2}$$

Put $x+2=0 \Rightarrow x=-2$ in (ii)

$$(-2)^3 + 22(-2)^2 + 14(-2) - 17 = D(-2-3)$$

$$-8 + 88 - 28 - 17 = D(-5)$$

$$35 = -5D \Rightarrow \boxed{D = -7}$$

Expanding (ii)

$$x^3 + 22x^2 + 14x - 17 = A(x^3 + 6x^2 + 12x + 8) + B(x - 3)(x^2 + 4x + 4) \\ + C(x^2 + 2x - 3x - 6) + D(x - 3)$$

$$x^3 + 22x^2 + 14x - 17 = A(x^3 + 6x^2 + 12x + 8) + B(x^3 + 4x^2 + 4x - 3x^2 \\ - 12x - 12) + C(x^2 - x - 6) + D(x - 3)$$

$$x^3 + 22x^2 + 14x - 17 = Ax^3 + 6Ax^2 + 12Ax + 8A + Bx^3 + Bx^2 - 8Bx \\ - 12B + Cx^2 - Cx - 6C + Dx - 3D$$

Comparing the coefficients of x^3

$$1 = A + B \dots \text{(iii)}$$

Put $A = 2$

$$1 = 2 + B \Rightarrow \boxed{B = -1}$$

Comparing the coefficients of x^2

$$22 = 6A + B + C$$

Put $A = 2$ and $B = -1$

$$22 = 6(2) + (-1) + C$$

$$22 = 12 - 1 + C$$

$$22 = 11 + C \Rightarrow \boxed{C = 11}$$

Put $A = 2, B = -1, C = 11, D = -7$ in (i)

$$\frac{x^3 + 22x^2 + 14x - 17}{(x - 3)(x + 2)^2} = \frac{2}{x - 3} - \frac{1}{x + 2} + \frac{11}{(x + 2)^2} - \frac{7}{(x + 2)^3}$$

Integrating both sides

$$\int \frac{x^3 + 22x^2 + 14x - 17}{(x - 3)(x + 2)^2} dx = 2 \int \frac{1}{x - 3} dx - \int \frac{1}{x + 2} dx + 11 \int (x + 2)^{-2} dx - 7 \int (x + 2)^{-3} dx \\ = 2 \ln|x - 3| - \ln|x + 2| + 11 \left(\frac{(x + 2)^{-2+1}}{-2+1} \right) - 7 \left(\frac{(x + 2)^{-3+1}}{-3+1} \right) + c \\ = 2 \ln|x - 3| - \ln|x + 2| - \frac{11}{x + 2} + \frac{7}{2(x + 2)^2} + c$$

Q.18 $\int \frac{x - 2}{(x + 1)(x^2 + 1)} dx$

Solution:

$$\text{Let } \frac{x - 2}{(x + 1)(x^2 + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1} \dots \text{(i)}$$

Multiplying both sides by $(x + 1)(x^2 + 1)$

$$x - 2 = A(x^2 + 1) + (Bx + C)(x + 1) \dots \text{(ii)}$$

Put $x + 1 = 0 \Rightarrow x = -1$ in (ii)

$$-1 - 2 = A\{(-1)^2 + 1\}$$

$$-3 = A(2) \Rightarrow \boxed{A = -\frac{3}{2}}$$

Expanding (ii)

$$x-2 = Ax^2 + A + Bx^2 + Bx + Cx + C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = \frac{-3}{2} + B \Rightarrow \boxed{B = \frac{3}{2}}$$

Comparing the coefficients of x

$$1 = B + C$$

$$1 = \frac{3}{2} + C$$

$$C = 1 - \frac{3}{2} \Rightarrow \boxed{C = -\frac{1}{2}}$$

Put $A = -\frac{3}{2}$, $B = \frac{3}{2}$ and $C = -\frac{1}{2}$ in (i)

$$\frac{x-2}{(x+1)(x^2+1)} = \frac{\left(\frac{-3}{2}\right)}{x+1} + \frac{\left(\frac{3x}{2} - \frac{1}{2}\right)}{x^2+1}$$

$$\begin{aligned} \frac{x-2}{(x+1)(x^2+1)} &= \frac{-3}{2(x+1)} + \frac{3x-1}{x^2+1} \\ &= \frac{-3}{2(x+1)} + \frac{3x-1}{2(x^2+1)} \end{aligned}$$

$$\frac{x-2}{(x+1)(x^2+1)} = \frac{-3}{2(x+1)} + \frac{3x}{2(x^2+1)} - \frac{1}{2(x^2+1)}$$

Integrating both sides

$$\begin{aligned} \int \frac{x-2}{(x+1)(x^2+1)} dx &= \frac{-3}{2} \int \frac{1}{x+1} dx + \frac{3}{2} \int \frac{x}{x^2+1} dx - \frac{1}{2} \int \frac{1}{x^2+1} dx \\ &= \frac{-3}{2} \int \frac{1}{x+1} dx + \frac{3}{4} \int \frac{2x}{x^2+1} dx - \frac{1}{2} \int \frac{1}{x^2+1} dx \\ &= \frac{-3}{2} \ln|x+1| + \frac{3}{4} \ln|x^2+1| - \frac{1}{2} \tan^{-1} x + c \end{aligned}$$

Q.19 $\int \frac{x}{(x-1)(x^2+1)} dx$

Solution:

$$\text{Let } \frac{x}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} \dots \text{ (i)}$$

Multiplying both sides by $(x-1)(x^2+1)$

$$x = A(x^2+1) + (Bx+C)(x-1) \dots \text{ (ii)}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$1 = A\{(1)^2+1\}$$

$$1 = A(2) \Rightarrow \boxed{A = \frac{1}{2}}$$

Expanding (ii)

$$x = Ax^2 + A + Bx^2 - Bx + Cx - C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = \frac{1}{2} + B \Rightarrow \boxed{B = -\frac{1}{2}}$$

Comparing the coefficients of “ x ”

$$1 = -B + C$$

$$1 = -\left(-\frac{1}{2}\right) + C$$

$$C = 1 - \frac{1}{2} \Rightarrow \boxed{C = \frac{1}{2}}$$

Put $A = \frac{1}{2}$, $B = -\frac{1}{2}$ and $C = \frac{1}{2}$ in (i)

$$\frac{x}{(x-1)(x^2+1)} = \frac{\left(\frac{1}{2}\right)}{x-1} + \frac{\left(-\frac{x}{2} + \frac{1}{2}\right)}{x^2+1}$$

$$\frac{x}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} + \frac{-x+1}{2(x^2+1)}$$

$$\frac{x}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{x}{2(x^2+1)} + \frac{1}{2(x^2+1)}$$

Integrating both sides

$$\begin{aligned} \int \frac{x}{(x-1)(x^2+1)} dx &= \frac{1}{2} \int \frac{1}{x-1} dx - \frac{1}{2} \int \frac{x}{x^2+1} dx + \frac{1}{2} \int \frac{1}{x^2+1} dx \\ &= \frac{1}{2} \int \frac{1}{x-1} dx - \frac{1}{2 \times 2} \int \frac{2x}{x^2+1} dx + \frac{1}{2} \int \frac{1}{x^2+1} dx \\ &= \frac{1}{2} \ln|x-1| - \frac{1}{4} \ln|x^2+1| + \frac{1}{2} \tan^{-1} x + c \end{aligned}$$

Q.20 $\int \frac{9x-7}{(x+3)(x^2+1)} dx$

Solution:

$$\text{Let } \frac{9x-7}{(x+3)(x^2+1)} = \frac{A}{x+3} + \frac{Bx+C}{x^2+1} \dots \text{(i)}$$

Multiplying both sides by $(x+3)(x^2+1)$

$$9x-7 = A(x^2+1) + (Bx+C)(x+3) \dots \text{(ii)}$$

Put $x+3=0 \Rightarrow x=-3$ in (ii)

$$9(-3)-7 = A\{(-3)^2+1\}$$

$$-34 = 10A \Rightarrow \boxed{A = \frac{-17}{5}}$$

Expanding (ii)

$$9x - 7 = Ax^2 + A + Bx^2 + 3Bx + Cx + 3C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = \frac{-17}{5} + B \Rightarrow \boxed{B = \frac{17}{5}}$$

Comparing the coefficients of x

$$9 = 3B + C$$

$$9 = 3\left(\frac{17}{5}\right) + C$$

$$C = 9 - \frac{51}{5}$$

$$C = \frac{45 - 51}{5} \Rightarrow \boxed{C = \frac{-6}{5}}$$

Put $A = \frac{-17}{5}, B = \frac{17}{5}, C = \frac{-6}{5}$ in (i)

$$\frac{9x - 7}{(x+3)(x^2+1)} = \frac{\frac{-17}{5}}{x+3} + \frac{\frac{17x}{5} + \left(\frac{-6}{5}\right)}{x^2+1}$$

$$= \frac{-17}{5(x+3)} + \frac{\frac{17x-6}{5}}{x^2+1}$$

$$= \frac{-17}{5(x+3)} + \frac{17x-6}{5(x^2+1)}$$

$$\frac{9x-7}{(x+3)(x^2+1)} = \frac{-17}{5(x+3)} + \frac{17x}{5(x^2+1)} - \frac{6}{5(x^2+1)}$$

Integrating both sides

$$\begin{aligned} \int \frac{9x-7}{(x+3)(x^2+1)} dx &= \frac{-17}{5} \int \frac{1}{x+3} dx + \frac{17}{5} \int \frac{x}{x^2+1} dx - \frac{6}{5} \int \frac{1}{x^2+1} dx \\ &= \frac{-17}{5} \int \frac{1}{x+3} dx + \frac{17}{5 \times 2} \int \frac{2x}{x^2+1} dx - \frac{6}{5} \int \frac{1}{x^2+1} dx \\ &= \frac{-17}{5} \ln|x+3| + \frac{17}{10} \ln|x^2+1| - \frac{6}{5} \tan^{-1} x + c \end{aligned}$$

Q.21 $\int \frac{1+4x}{(x-3)(x^2+4)} dx$

Solution:

Let $\frac{1+4x}{(x-3)(x^2+4)} = \frac{A}{x-3} + \frac{Bx+C}{x^2+4} \dots$ (i)

Multiplying both sides by $(x-3)(x^2+4)$

$$1+4x = A(x^2+4) + (Bx+C)(x-3) \dots (ii)$$

Put $x-3=0 \Rightarrow x=3$ in (ii)

$$1+4(3) = A\{(3)^2+4\}$$

$$13 = 13A \Rightarrow \boxed{A=1}$$

Expanding (ii)

$$1+4x = Ax^2 + 4A + Bx^2 - 3Bx + Cx - 3C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = 1 + B \Rightarrow \boxed{B=-1}$$

Comparing the coefficients of 'x'

$$4 = -3B + C$$

$$4 = -3(-1) + C$$

$$4 = 3 + C \Rightarrow \boxed{C=1}$$

Put $A=1, B=-1, C=1$ in (i)

$$\frac{1+4x}{(x-3)(x^2+4)} = \frac{1}{x-3} + \frac{(-1)x+1}{x^2+4}$$

$$\frac{1+4x}{(x-3)(x^2+4)} = \frac{1}{x-3} + \frac{-x}{x^2+4} + \frac{1}{x^2+4}$$

Integrating both sides

$$\begin{aligned} \int \frac{1+4x}{(x-3)(x^2+4)} dx &= \int \frac{1}{x-3} dx + \frac{-1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{1}{x^2+(2)^2} dx \\ &= \ln|x-3| - \frac{1}{2} \ln|x^2+4| + \frac{1}{2} \tan^{-1} \frac{x}{2} + c \end{aligned}$$

Q.22 $\int \frac{12}{x^3+8} dx$

Solution:

$$\frac{12}{(x^3+8)} = \frac{12}{(x)^3+(2)^3} = \frac{12}{(x+2)(x^2-2x+4)}$$

$$\text{Let } \frac{12}{(x+2)(x^2-2x+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2-2x+4} \dots (i)$$

Multiplying both sides by $(x+2)(x^2-2x+4)$

$$12 = A(x^2-2x+4) + (Bx+C)(x+2) \dots (ii)$$

Put $x+2=0 \Rightarrow x=-2$ in (ii)

$$12 = A\{(-2)^2 - 2(-2) + 4\}$$

$$12 = A(4+4+4)$$

$$12 = 12A \Rightarrow \boxed{A=1}$$

Expanding (ii)

$$12 = Ax^2 - 2Ax + 4A + Bx^2 + 2Bx + Cx + 2C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = 1 + B \Rightarrow \boxed{B = -1}$$

Comparing the coefficients of 'x'

$$0 = -2A + 2B + C$$

$$0 = -2(1) + 2(-1) + C$$

$$0 = -4 + C \Rightarrow \boxed{C = 4}$$

Put $A = 1, B = -1, C = 4$ in (i)

$$\frac{12}{(x+2)(x^2-2x+4)} = \frac{1}{x+2} + \frac{(-1)x+4}{x^2-2x+4}$$

$$\frac{12}{x^3+8} = \frac{1}{x+2} - \frac{x-4}{x^2-2x+4}$$

$$= \frac{1}{x+2} - \frac{x-1-3}{x^2-2x+4}$$

$$\frac{12}{x^3+8} = \frac{1}{x+2} - \frac{x-1}{x^2-2x+4} + \frac{3}{x^2-2x+4}$$

Integrating both sides

$$\begin{aligned} \int \frac{12}{x^3+8} dx &= \int \frac{1}{x+2} dx - \int \frac{x-1}{x^2-2x+4} dx + 3 \int \frac{1}{x^2-2x+4} dx \\ &= \int \frac{1}{x+2} dx - \frac{1}{2} \int \frac{2x-2}{x^2-2x+4} dx + 3 \int \frac{1}{(x-1)^2+(\sqrt{3})^2} dx \\ &= \ln|x+2| - \frac{1}{2} \ln|x^2-2x+4| + 3 \left(\frac{1}{\sqrt{3}} \tan^{-1} \frac{x-1}{\sqrt{3}} \right) + c \\ &= \ln|x+2| - \frac{1}{2} \ln|x^2-2x+4| + \sqrt{3} \tan^{-1} \left(\frac{x-1}{\sqrt{3}} \right) + c \end{aligned}$$

Q.23 $\int \frac{9x+6}{x^3-8} dx$

Solution:

Consider $\frac{9x+6}{x^3-8} = \frac{9x+6}{(x)^3-(2)^3} = \frac{9x+6}{(x-2)(x^2+2x+4)}$

Let $\frac{9x+6}{(x-2)(x^2+2x+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4} \dots$ (i)

Multiplying both sides by $(x-2)(x^2+2x+4)$

$$9x+6 = A(x^2+2x+4) + (Bx+C)(x-2) \dots$$
 (ii)

Put $x-2=0 \Rightarrow x=2$ in (ii)

$$9(2)+6 = A\{(2)^2+2(2)+4\}$$

$$24 = A(12) \Rightarrow \boxed{A=2}$$

Expanding (ii)

$$9x+6 = Ax^2+2Ax+4A+Bx^2-2Bx+Cx-2C$$

Comparing the coefficients of x^2

$$0 = A + B$$

$$0 = 2 + B \Rightarrow \boxed{B = -2}$$

Comparing the coefficients of x

$$9 = 2A - 2B + C$$

$$9 = 2(2) - 2(-2) + C$$

$$9 = 4 + 4 + C \Rightarrow \boxed{C = 1}$$

Put $A = 2, B = -2$ and $C = 1$ in (i)

$$\begin{aligned} \frac{9x+6}{(x-2)(x^2+2x+4)} &= \frac{2}{x-2} + \frac{(-2)x+1}{x^2+2x+4} \\ \frac{9x+6}{x^3-8} &= \frac{2}{x-2} - \frac{2x-1}{x^2+2x+4} \\ &= \frac{2}{x-2} - \frac{2x+2-2-1}{x^2+2x+4} \\ \frac{9x+6}{x^3-8} &= \frac{2}{x-2} - \frac{2x+2}{x^2+2x+4} + \frac{3}{x^2+2x+4} \end{aligned}$$

Integrating both sides

$$\begin{aligned} \int \frac{9x+6}{x^3-8} dx &= 2 \int \frac{1}{x-2} dx - \int \frac{2x+2}{x^2+2x+4} dx + 3 \int \frac{1}{x^2+2x+1+3} dx \\ &= 2 \ln|x-2| - \ln|x^2+2x+4| + 3 \int \frac{1}{(x+1)^2 + (\sqrt{3})^2} dx \\ &= 2 \ln|x-2| - \ln|x^2+2x+4| + 3 \left(\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+1}{\sqrt{3}} \right) \right) + c \\ &= 2 \ln|x-2| - \ln|x^2+2x+4| + \sqrt{3} \tan^{-1} \left(\frac{x+1}{\sqrt{3}} \right) + c \end{aligned}$$

Q.24 $\int \frac{2x^2+5x+3}{(x-1)^2(x^2+4)} dx$

Solution:

$$\text{Let } \frac{2x^2+5x+3}{(x-1)^2(x^2+4)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+4} \dots \text{ (i)}$$

Multiplying both sides by $(x-1)^2(x^2+4)$

$$2x^2+5x+3 = A(x-1)(x^2+4) + B(x^2+4) + (Cx+D)(x-1)^2 \dots \text{ (ii)}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$2(1)^2+5(1)+3 = B\{(1)^2+4\}$$

$$10 = 5B \Rightarrow \boxed{B = 2}$$

Expanding (ii)

$$2x^2+5x+3 = A(x^3+4x-x^2-4) + B(x^2+4) + (Cx+D)(x^2+1-2x)$$

$$2x^2+5x+3 = Ax^3 - Ax^2 + 4Ax - 4A + Bx^2 + 4B + Cx^3 + Cx - 2Cx^2 + Dx^2 + D - 2Dx$$

Comparing the coefficients of x^3

$$0 = A + C \dots \text{ (iii)}$$

Comparing the coefficients of x^2

$$2 = -A + B - 2C + D$$

Put $B = 2$ and $A = -C$ (from (iii))

$$2 = C + 2 - 2C + D$$

$$0 = -C + D$$

$$C = D \dots \text{(iv)}$$

Comparing the coefficients of ' x '

$$5 = 4A + C - 2D$$

Put $D = C$ and $A = -C$

$$5 = 4(-C) + C - 2C$$

$$5 = -5C \Rightarrow \boxed{C = -1}$$

From (iv) $\boxed{D = -1}$

Put $C = -1$ in (iii)

$$0 = A + (-1) \Rightarrow \boxed{A = 1}$$

Put $A = 1, B = 2, C = -1, D = -1$ in (i)

$$\frac{2x^2 + 5x + 3}{(x-1)^2(x^2 + 4)} = \frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{(-1)x + (-1)}{x^2 + 4}$$

$$\frac{2x^2 + 5x + 3}{(x-1)^2(x^2 + 4)} = \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{x}{x^2 + 4} - \frac{1}{x^2 + 4}$$

Integrating both sides

$$\begin{aligned} \int \frac{2x^2 + 5x + 3}{(x-1)^2(x^2 + 4)} dx &= \int \frac{1}{x-1} dx + 2 \int (x-1)^{-2} dx - \frac{1}{2} \int \frac{2x}{x^2 + 4} dx - \int \frac{1}{x^2 + 2^2} dx \\ &= \ln|x-1| + 2 \left(\frac{(x-1)^{-2+1}}{-2+1} \right) - \frac{1}{2} \ln|x^2 + 4| - \frac{1}{2} \tan^{-1} \frac{x}{2} + c \\ &= \ln|x-1| - \frac{2}{x-1} - \frac{1}{2} \ln|x^2 + 4| - \frac{1}{2} \tan^{-1} \frac{x}{2} + c \end{aligned}$$

Q.25 $\int \frac{2x^2 - x - 7}{(x+2)^2(x^2 + x + 1)} dx$

Solution:

$$\text{Let } \frac{2x^2 - x - 7}{(x+2)^2(x^2 + x + 1)} = \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{Cx + D}{x^2 + x + 1} \dots \text{(i)}$$

Multiplying both sides by $(x+2)^2(x^2 + x + 1)$

$$2x^2 - x - 7 = A(x+2)(x^2 + x + 1) + B(x^2 + x + 1) + (Cx + D)(x+2)^2 \dots \text{(ii)}$$

Put $x + 2 = 0 \Rightarrow x = -2$ in (ii)

$$2(-2)^2 - (-2) - 7 = B\{(-2)^2 + (-2) + 1\}$$

$$2(4) + 2 - 7 = B\{4 - 1\}$$

$$3 = 3B \Rightarrow \boxed{B = 1}$$

Expanding (ii)

$$2x^2 - x - 7 = A(x^3 + x^2 + x + 2x^2 + 2x + 2) + B(x^2 + x + 1) + (Cx + D)(x^2 + 4x + 4)$$

$$2x^2 - x - 7 = Ax^3 + 3Ax^2 + 3Ax + 2A + Bx^2 + Bx + B + Cx^3 + 4Cx^2 + 4Cx + Dx^2 + 4Dx + 4D$$

Comparing the coefficients of x^3

$$0 = A + C \dots \text{(iii)}$$

Comparing the coefficients of x^2

$$2 = 3A + B + 4C + D$$

Put $A = -C$ (from (iii))

$$2 = -3C + B + 4C + D$$

$$2 = B + C + D \dots \text{(iv)}$$

Comparing the coefficients of x

$$-1 = 3A + B + 4C + 4D$$

$$-1 = 3A + 3C + B + C + D + 3D$$

$$-1 = 3(A + C) + (B + C + D) + 3D$$

$$-1 = 3(0) + 2 + 3D \quad (A + C = 0, B + C + D = 2)$$

$$3D = -1 - 2 \Rightarrow \boxed{D = -1}$$

Put $B = 1, D = -1$ in (iv)

$$2 = 1 + C + (-1) \Rightarrow \boxed{C = 2}$$

Put $C = 2$ in (iii)

$$0 = A + 2 \Rightarrow \boxed{A = -2}$$

Put $A = -2, B = 1, C = 2, D = -1$ in (i)

$$\frac{2x^2 - x - 7}{(x+2)^2(x^2+x+1)} = -\frac{2}{x+2} + \frac{1}{(x+2)^2} + \frac{2x-1}{x^2+x+1}$$

Integrating both sides

$$\begin{aligned} \int \frac{2x^2 - x - 7}{(x+2)^2(x^2+x+1)} dx &= -2 \int \frac{dx}{x+2} + \int (x+2)^{-2} dx + \int \frac{2x-1}{x^2+x+1} dx \\ &= -2 \ln|x+2| + \frac{(x+2)^{-1}}{-1} + \int \frac{2x+1-2}{x^2+x+1} dx \\ &= -2 \ln|x+2| - \frac{1}{x+2} + \int \frac{2x+1}{x^2+x+1} dx - 2 \int \frac{dx}{x^2+x+1} \\ &= -2 \ln|x+2| - \frac{1}{x+2} + \ln|x^2+x+1| - 2 \int \frac{dx}{x^2+2x+\frac{1}{2}+\left(\frac{1}{2}\right)^2-\frac{1}{4}+1} \\ &= -2 \ln|x+2| - \frac{1}{x+2} + \ln|x^2+x+1| - 2 \int \frac{dx}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= -2 \ln|x+2| - \frac{1}{x+2} + \ln|x^2+x+1| - 2 \left[\frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right] + c \\ &= -2 \ln|x+2| - \frac{1}{x+2} + \ln|x^2+x+1| - \frac{4}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + c \end{aligned}$$

Q.26 $\int \frac{3x+1}{(4x^2+1)(x^2-x+1)} dx$

Solution:

Let $\frac{3x+1}{(4x^2+1)(x^2-x+1)} = \frac{Ax+B}{4x^2+1} + \frac{Cx+D}{x^2-x+1} \dots$ (i)

Multiplying both sides by $(4x^2+1)(x^2-x+1)$

$$3x+1 = (Ax+B)(x^2-x+1) + (Cx+D)(4x^2+1)$$

$$3x+1 = Ax^3 - Ax^2 + Ax + Bx^2 - Bx + B + 4Cx^3 + Cx + 4Dx^2 + D$$

Comparing coefficients of like powers of x , we have

(for x^3) $A+4C=0 \dots$ (ii)

(for x^2) $-A+B+4D=0 \dots$ (iii)

(for x) $A-B+C=3 \dots$ (iv)

(Constant) $B+D=1$

$B=1-D \dots$ (v)

Adding (iii) and (iv)

$$4D+C=3$$

$C=3-4D \dots$ (vi)

From (ii) $A=-4C \dots$ (vii)

So $A=-4(3-4D)$

$A=-12+16D \dots$ (viii)

Put values of A , B and C in (iv)

$$-12+16D-(1-D)+3-4D=3$$

$$-12+16D-1+D+3-4D=3$$

$$13D-10=3$$

$$13D=13 \Rightarrow \boxed{D=1}$$

From (v) $B=1-1 \Rightarrow \boxed{B=0}$

From (vi) $C=3-4(1) \Rightarrow \boxed{C=-1}$

From (viii) $A=-12+16(1) \Rightarrow \boxed{A=4}$

Putting values of A , B , C and D in (i)

$$\begin{aligned} \frac{3x+1}{(4x^2+1)(x^2-x+1)} &= \frac{4x+0}{4x^2+1} + \frac{-x+1}{x^2-x+1} \\ &= \frac{4x}{4x^2+1} - \frac{x-1}{x^2-x+1} \end{aligned}$$

Integrating both sides

$$\begin{aligned} \int \frac{3x+1}{(4x^2+1)(x^2-x+1)} dx &= \frac{1}{2} \int \frac{8x}{4x^2+1} dx - \frac{1}{2} \int \frac{2x-2}{x^2-x+1} dx \\ &= \frac{1}{2} \ln(4x^2+1) - \frac{1}{2} \int \frac{2x-1-1}{x^2-x+1} dx \\ &= \frac{1}{2} \ln(4x^2+1) - \frac{1}{2} \int \frac{2x-1}{x^2-x+1} dx + \frac{1}{2} \int \frac{1}{x^2-x+1} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{2} \int \frac{1}{x^2 - x + \left(\frac{1}{2}\right)^2 - \frac{1}{4} + 1} dx \\
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{2} \int \frac{1}{x^2 - x + \left(\frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{2} \int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx \\
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{2} \frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1} \left(\frac{x - \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + c \\
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{2} \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\frac{2x-1}{2}}{\frac{\sqrt{3}}{2}} \right) + c \\
&= \frac{1}{2} \ln(4x^2 + 1) - \frac{1}{2} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) + c
\end{aligned}$$

Q.27 $\int \frac{4x+1}{(x^2+4)(x^2+4x+5)} dx$

Solution:

Let $\frac{4x+1}{(x^2+4)(x^2+4x+5)} = \frac{Ax+B}{x^2+4} + \frac{Cx+D}{x^2+4x+5} \dots$ (i)

Multiplying both sides by $(x^2+4)(x^2+4x+5)$

$$4x+1 = (Ax+B)(x^2+4x+5) + (Cx+D)(x^2+4)$$

$$4x+1 = Ax^3 + 4Ax^2 + 5Ax + Bx^2 + 4Bx + 5B + Cx^3 + 4Cx + Dx^2 + 4D$$

Comparing coefficients of like powers of x .

(For x^3) $A+C=0 \dots$ (ii)

(For x^2) $4A+B+D=0 \dots$ (iii)

(For x) $5A+4B+4C=4 \dots$ (iv)

(For constant) $5B+4D=1 \dots$ (v)

From (ii) $A=-C \dots$ (vi)

Put $A=-C$ in (iv)

$$-5C+4B+4C=4$$

$$4B-C=4 \dots$$
 (vii)

From (v) $D = \frac{1-5B}{4} \dots$ (viii)

Putting values A and D in (iii)

$$4(-C) + B + \frac{1-5B}{4} = 0$$

$$\frac{-16C + 4B + 1 - 5B}{4} = 0$$

$$-16C - B + 1 = 0$$

$$B + 16C = 1 \dots (ix)$$

By Eq (vii) -4Eq (ix)

$$4B - C = 4$$

$$\pm 4B \pm 64C = \pm 4$$

$$-65C = 0$$

$$\boxed{C = 0}$$

$$\text{From (vii) } 4B - 0 = 4 \Rightarrow \boxed{B = 1}$$

$$\text{From (vi) } \boxed{A = 0}$$

$$\text{From (viii) } D = \frac{1-5}{4} \Rightarrow \boxed{D = -1}$$

Putting values of A, B, C and D in (i)

$$\frac{4x+1}{(x^2+4)(x^2+4x+5)} = \frac{1}{x^2+4} - \frac{1}{x^2+4x+5}$$

Integrating both sides

$$\begin{aligned} \int \frac{4x+1}{(x^2+4)(x^2+4x+5)} dx &= \int \frac{1}{x^2+4} dx - \int \frac{1}{x^2+4x+5} dx \\ &= \int \frac{1}{(x)^2 + (2)^2} dx - \int \frac{1}{(x+2)^2 + (1)^2} dx \\ &= \frac{1}{2} \tan^{-1} \frac{x}{2} - \tan^{-1}(x+2) + c \end{aligned}$$

Q.28 $\int \frac{6a^2}{(x^2+a^2)(x^2+4a^2)} dx$

Solution:

$$\text{Let } \frac{6a^2}{(x^2+a^2)(x^2+4a^2)} = \frac{Ax+B}{x^2+a^2} + \frac{Cx+D}{x^2+4a^2} \dots (i)$$

Multiplying both sides by $(x^2+a^2)(x^2+4a^2)$

$$6a^2 = (Ax+B)(x^2+4a^2) + (Cx+D)(x^2+a^2)$$

$$6a^2 = Ax^3 + 4a^2Ax + Bx^2 + 4a^2B + Cx^3 + a^2Cx + Dx^2 + a^2D$$

Comparing coefficients of like powers of x, we have

$$\text{(For } x^3) \quad A+C=0 \dots (ii)$$

$$\text{(For } x^2) \quad B+D=0 \dots (iii)$$

$$\text{(For } x) \quad 4a^2A+a^2C=0 \dots (iv)$$

$$\text{(For constant) } 4a^2B+a^2D=6a^2 \dots (v)$$

$$\text{From (ii) } A=-C \dots (vi)$$

From (iv) $4a^2A = -a^2C \Rightarrow A = -\frac{1}{4}C \dots$ (vii)

From (vi) and (vii)

$$-C = -\frac{1}{4}C \Rightarrow \frac{1}{4}C - C = 0$$

$$-\frac{3}{4}C = 0 \Rightarrow \boxed{C = 0}$$

From (ii) $\boxed{A = 0}$

From (v) $4a^2B + a^2D = 6a^2$

$$4B + D = 6 \dots$$
 (viii)

By Eq (viii) – Eq (iii)

$$4B + D = 6$$

$$\pm B \pm D = 0$$

$$3B = 6$$

$$\Rightarrow \boxed{B = 2}$$

From (iii) $2 + D = 0 \Rightarrow \boxed{D = -2}$

Putting values of A, B, C , and D in eq. (i)

$$\frac{6a^2}{(x^2 + a^2)(x^2 + 4a^2)} = \frac{2}{x^2 + a^2} - \frac{2}{x^2 + 4a^2}$$

Integrating both sides

$$\begin{aligned} \int \frac{6a^2}{(x^2 + a^2)(x^2 + 4a^2)} dx &= 2 \int \frac{1}{x^2 + a^2} dx - 2 \int \frac{1}{x^2 + (2a)^2} dx \\ &= 2 \left(\frac{1}{a} \right) \tan^{-1} \frac{x}{a} - 2 \left(\frac{1}{2a} \right) \tan^{-1} \frac{x}{2a} + c \\ &= \frac{2}{a} \tan^{-1} \frac{x}{a} - \frac{1}{a} \tan^{-1} \frac{x}{2a} + c \end{aligned}$$

Q.29 $\int \frac{2x^2 - 2}{x^4 + x^2 + 1} dx$

Solution:

$$\begin{aligned} \frac{2x^2 - 2}{x^4 + x^2 + 1} &= \frac{2x^2 - 2}{x^4 + 1 + 2x^2 - 2x^2 + x^2} \\ &= \frac{2x^2 - 2}{(x^2 + 1)^2 - (x)^2} = \frac{2x^2 - 2}{(x^2 + 1 + x)(x^2 + 1 - x)} \end{aligned}$$

Let $\frac{2x^2 - 2}{(x^2 + x + 1)(x^2 - x + 1)} = \frac{Ax + B}{x^2 + x + 1} + \frac{Cx + D}{x^2 - x + 1} \dots$ (i)

Multiplying both sides by $(x^2 + x + 1)(x^2 - x + 1)$

$$2x^2 - 2 = (Ax + B)(x^2 - x + 1) + (Cx + D)(x^2 + x + 1)$$

$$2x^2 - 2 = Ax^3 - Ax^2 + Ax + Bx^2 - Bx + B + Cx^3 + Cx^2 + Cx + Dx^2 + Dx + D$$

Comparing coefficients of like powers of x ,

(For x^3) $A + C = 0 \dots$ (ii)

(For x^2) $-A + B + C + D = 2 \dots$ (iii)

(For x) $A - B + C + D = 0 \dots$ (iv)

(For constant) $B + D = -2 \dots$ (v)

Put $A = -C$ (From (ii)) in (iv)

$$-C - B + C + D = 0$$

$$-B + D = 0 \dots$$
 (vi)

By Eq (vi) + Eq (v)

$$-B + D = 0$$

$$\begin{array}{r} B + D = -2 \\ \hline 2D = -2 \end{array}$$

$$\boxed{D = -1}$$

From (v) $B - 1 = -2 \Rightarrow \boxed{B = -1}$

Put values of B and D in (iii)

$$-A - 1 + C - 1 = 2$$

$$-A + C = 4 \dots$$
 (vii)

By Eq (ii) + Eq (vii)

$$A + C = 0$$

$$\begin{array}{r} -A + C = 4 \\ \hline 2C = 4 \end{array}$$

$$\boxed{C = 2}$$

From (ii) $\boxed{A = -2}$

Putting values of A, B, C and D in (i)

$$\frac{2x^2 - 2}{(x^2 + x + 1)(x^2 - x + 1)} = \frac{-2x - 1}{x^2 + x + 1} + \frac{2x - 1}{x^2 - x + 1}$$

$$\frac{2x^2 - 2}{x^4 + x^2 + 1} = -\frac{2x + 1}{x^2 + x + 1} + \frac{2x - 1}{x^2 - x + 1}$$

Integrating both sides

$$\begin{aligned} \int \frac{2x^2 - 2}{x^4 + x^2 + 1} dx &= -\int \frac{2x + 1}{x^2 + x + 1} dx + \int \frac{2x - 1}{x^2 - x + 1} dx \\ &= -\ln(x^2 + x + 1) + \ln(x^2 - x + 1) + c \\ &= \ln\left(\frac{x^2 - x + 1}{x^2 + x + 1}\right) + c \end{aligned}$$

Q.30 $\int \frac{3x - 8}{(x^2 - x + 2)(x^2 + x + 2)} dx$

Solution:

$$\text{Let } \frac{3x - 8}{(x^2 - x + 2)(x^2 + x + 2)} = \frac{Ax + B}{x^2 - x + 2} + \frac{Cx + D}{x^2 + x + 2} \dots \text{ (i)}$$

Multiplying both sides by $(x^2 - x + 2)(x^2 + x + 2)$

$$3x - 8 = (Ax + B)(x^2 + x + 2) + (Cx + D)(x^2 - x + 2)$$

$$3x - 8 = Ax^3 + Ax^2 + 2Ax + Bx^2 + Bx + 2B + Cx^3 - Cx^2 + 2Cx + Dx^2 - Dx + 2D$$

Comparing coefficients of like powers of x ,

$$\text{(For } x^3) \quad A + C = 0 \dots \text{(ii)}$$

$$\text{(For } x^2) \quad A + B - C + D = 0 \dots \text{(iii)}$$

$$\text{(For } x) \quad 2A + B + 2C - D = 3 \dots \text{(iv)}$$

$$\text{(For constant)} \quad 2B + 2D = -8 \Rightarrow B + D = -4 \dots \text{(v)}$$

By Eq (iii) - Eq (v)

$$A - C = 4 \dots \text{(vi)}$$

By Eq (ii) + Eq (vi)

$$A + C = 0$$

$$\frac{A - C = 4}{2A = 4}$$

$$2A = 4$$

$$\boxed{A = 2}$$

$$\text{From (ii)} \quad \boxed{C = -2}$$

$$\text{From (ii) and (iv)} \quad B - D = 3 \dots \text{(vii)}$$

By Eq (v) + Eq (vii)

$$B + D = -4$$

$$\frac{B - D = 3}{2B = -1}$$

$$\boxed{B = -\frac{1}{2}}$$

$$\text{From (v)} \quad \frac{-1}{2} + D = -4$$

$$D = -4 + \frac{1}{2} \Rightarrow \boxed{D = -\frac{7}{2}}$$

Putting values of A, B, C and D in (i)

$$\frac{3x - 8}{(x^2 - x + 2)(x^2 + x + 2)} = \frac{2x - \frac{1}{2}}{x^2 - x + 2} + \frac{-2x - \frac{7}{2}}{x^2 + x + 2}$$

Integrating both sides

$$\begin{aligned} \int \frac{3x - 8}{(x^2 - x + 2)(x^2 + x + 2)} dx &= \int \frac{2x - 1 + 1 - \frac{1}{2}}{x^2 - x + 2} dx - \int \frac{2x + 1 - 1 + \frac{7}{2}}{x^2 + x + 2} dx \\ &= \int \frac{2x - 1}{x^2 - x + 2} dx + \frac{1}{2} \int \frac{1}{x^2 - x + 2} dx - \int \frac{2x + 1}{x^2 + x + 2} dx - \frac{5}{2} \int \frac{1}{x^2 + x + 2} dx \end{aligned}$$

$$\begin{aligned}
&= \ln(x^2 - x + 2) + \frac{1}{2} \int \frac{1}{x^2 - x + \left(\frac{1}{2}\right)^2 - \frac{1}{4} + 2} dx - \ln(x^2 + x + 2) - \frac{5}{2} \int \frac{1}{x^2 + x + \frac{1}{4} + 2 - \frac{1}{4}} dx \\
&= \ln(x^2 - x + 2) + \frac{1}{2} \int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \frac{7}{4}} dx - \ln(x^2 + x + 2) - \frac{5}{2} \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}} dx \\
&= \ln(x^2 - x + 2) + \frac{1}{2} \int \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} dx - \ln(x^2 + x + 2) - \frac{5}{2} \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} dx \\
&= \ln(x^2 - x + 2) + \frac{1}{2} \left(\frac{1}{\left(\frac{\sqrt{7}}{2}\right)} \tan^{-1} \left(\frac{x - \frac{1}{2}}{\frac{\sqrt{7}}{2}} \right) \right) - \ln(x^2 + x + 2) - \frac{5}{2} \left(\frac{1}{\left(\frac{\sqrt{7}}{2}\right)} \tan^{-1} \left(\frac{x + \frac{1}{2}}{\frac{\sqrt{7}}{2}} \right) \right) + c \\
&= \ln(x^2 - x + 2) + \frac{1}{2} \times \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{2x - 1}{\frac{2}{\sqrt{7}}} \right) - \ln(x^2 + x + 2) - \frac{5}{2} \times \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{2x + 1}{\frac{2}{\sqrt{7}}} \right) + c \\
&= \ln(x^2 - x + 2) + \frac{1}{\sqrt{7}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{7}} \right) - \ln(x^2 + x + 2) - \frac{5}{\sqrt{7}} \tan^{-1} \left(\frac{2x + 1}{\sqrt{7}} \right) + c
\end{aligned}$$

Q.31 $\int \frac{3x^3 + 4x^2 + 9x + 5}{(x^2 + x + 1)(x^2 + 2x + 3)} dx$

Solution:

Let $\frac{3x^3 + 4x^2 + 9x + 5}{(x^2 + x + 1)(x^2 + 2x + 3)} = \frac{Ax + B}{x^2 + x + 1} + \frac{Cx + D}{x^2 + 2x + 3} \dots$ (i)

Multiplying both sides by $(x^2 + x + 1)(x^2 + 2x + 3)$

$$3x^3 + 4x^2 + 9x + 5 = (Ax + B)(x^2 + 2x + 3) + (Cx + D)(x^2 + x + 1)$$

$$3x^3 + 4x^2 + 9x + 5 = Ax^3 + 2Ax^2 + 3Ax + Bx^2 + 2Bx + 3B + Cx^3 + Cx^2 + Cx + Dx^2 + Dx + D$$

Comparing coefficients of like powers of x ,

(For x^3) $A + C = 3 \dots$ (ii)

(For x^2) $2A + B + C + D = 4 \dots$ (iii)

(For x) $3A + 2B + C + D = 9 \dots$ (iv)

(For constant) $3B + D = 5 \Rightarrow D = 5 - 3B \dots$ (v)

From (ii) $A = 3 - C$

Put values of A and D in (iii)

$$6 - 2C + B + C + 5 - 3B = 4$$

$$-C - 2B = -7 \Rightarrow C + 2B = 7 \dots \text{(vi)}$$

Put values of A and D in (iv)

$$9 - 3C + 2B + C + 5 - 3B = 9$$

$$-2C - B = -5 \Rightarrow 2C + B = 5 \dots \text{(vii)}$$

By Eq (vi) $-2 \times$ Eq (vii)

$$C + 2B = 7$$

$$\pm 4C \pm 2B = \pm 10$$

$$\hline -3C = -3$$

$$\boxed{C=1}$$

From (vii) $2 + B = 5 \Rightarrow \boxed{B=3}$

From (v) $D = 5 - 3(3) \Rightarrow \boxed{D=-4}$

Put $C = 1$ in (ii)

$$A + 1 = 3 \Rightarrow \boxed{A=2}$$

Putting values of A, B, C and D in (i)

$$\frac{3x^3 + 4x^2 + 9x + 5}{(x^2 + x + 1)(x^2 + 2x + 3)} = \frac{2x + 3}{x^2 + x + 1} + \frac{x - 4}{x^2 + 2x + 3}$$

Integrating both sides

$$\begin{aligned} \int \frac{3x^3 + 4x^2 + 9x + 5}{(x^2 + x + 1)(x^2 + 2x + 3)} dx &= \int \frac{2x + 3}{x^2 + x + 1} dx + \int \frac{x - 4}{x^2 + 2x + 3} dx \\ &= \int \frac{2x + 1 + 2}{x^2 + x + 1} dx + \frac{1}{2} \int \frac{2x + 2 - 10}{x^2 + 2x + 3} dx \\ &= \int \frac{2x + 1}{x^2 + x + 1} dx + 2 \int \frac{1}{x^2 + x + 1} dx + \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 3} dx - \frac{10}{2} \int \frac{1}{x^2 + 2x + 3} dx \\ &= \ln(x^2 + x + 1) + 2 \int \frac{1}{x^2 + x + \left(\frac{1}{2}\right)^2 + 1 - \frac{1}{4}} dx + \frac{1}{2} \ln(x^2 + 2x + 3) - 5 \int \frac{1}{x^2 + 2x + 1 + 2} dx \\ &= \ln(x^2 + x + 1) + 2 \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx + \frac{1}{2} \ln(x^2 + 2x + 3) - 5 \int \frac{1}{(x + 1)^2 + (\sqrt{2})^2} dx \end{aligned}$$

$$= \ln(x^2 + x + 1) + 2 \frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1} \left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + \frac{1}{2} \ln(x^2 + 2x + 3) - 5 \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c$$

$$= \ln(x^2 + x + 1) + \frac{4}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{1}{2} \ln(x^2 + 2x + 1) - \frac{5}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c$$

$$= \ln(x^2 + x + 1) + \ln(\sqrt{x^2 + 2x + 1}) + \frac{4}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) - \frac{5}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c$$

$$= \ln|(x^2 + x + 1)\sqrt{x^2 + 2x + 3}| + \frac{4}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) - \frac{5}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c$$